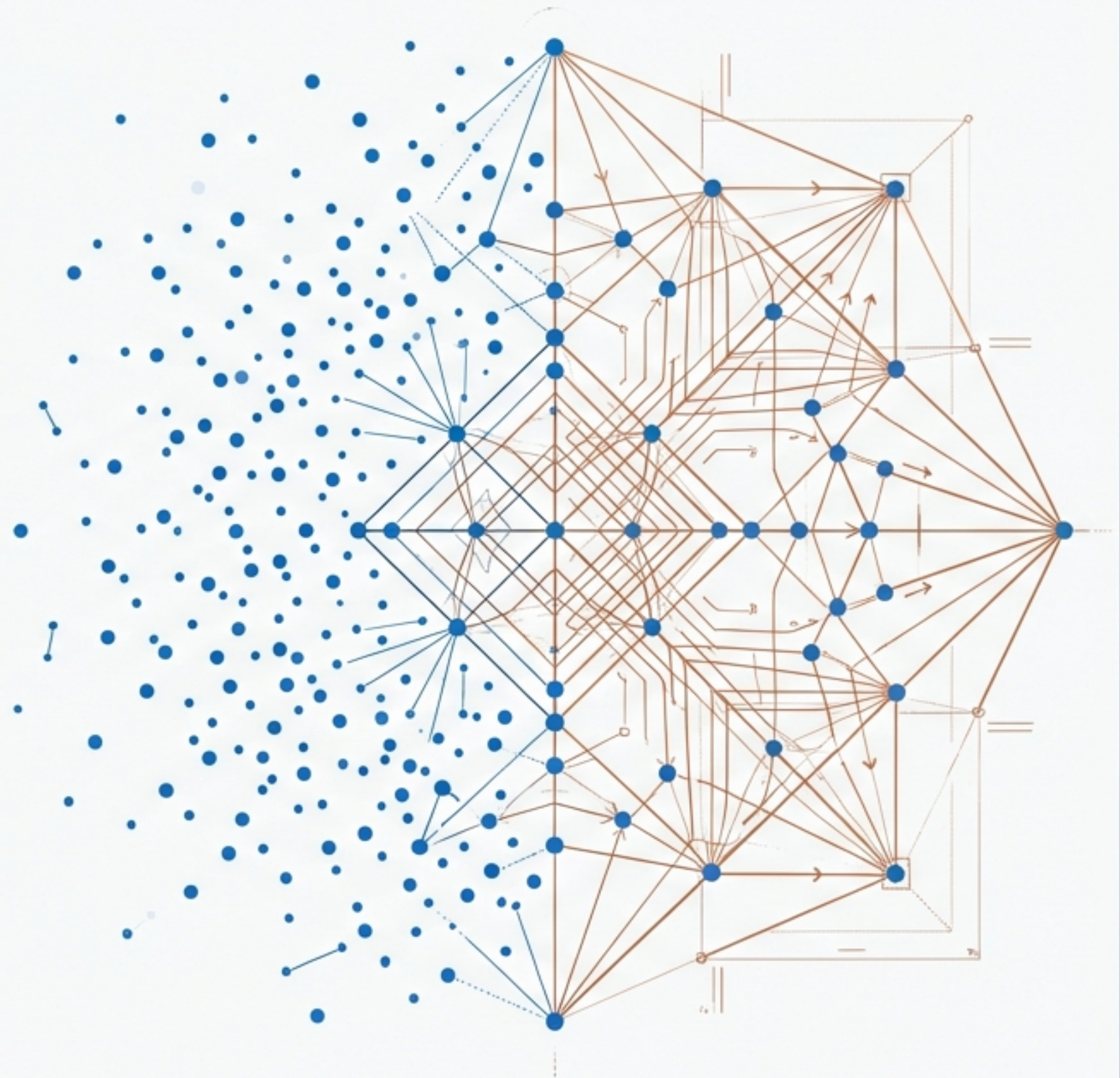


THE NEXT EVOLUTION OF ENTERPRISE AI: SCALING GRAPH-BASED RAG

Unlocking Multi-Hop Reasoning,
Auditability, and ROI at
Enterprise Scale



EXECUTIVE SUMMARY: BREAKING THE GENERATIVE AI CEILING

THE REALITY: Current Generative AI investments (Standard Vector RAG) are stalling on complex, highly-regulated enterprise data.

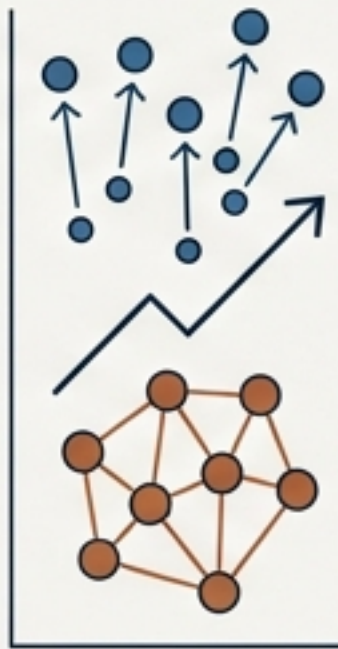
THE INTERVENTION: Graph-based Retrieval is the proven solution to unlock multi-hop reasoning, guarantee compliance, and reduce resolution times.

THE OBJECTIVE: Secure sponsorship to scale our hybrid, cost-optimized GraphRAG framework.

78.5%

WIN-RATE

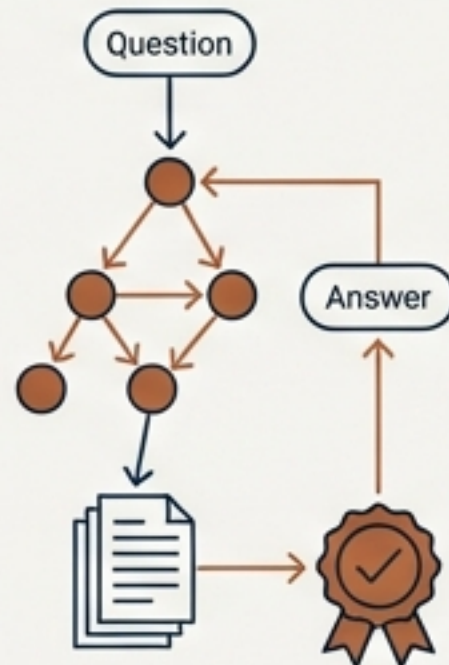
Unmatched precision over standard dense vector retrieval on complex legacy tasks.



100%

AUDITABILITY

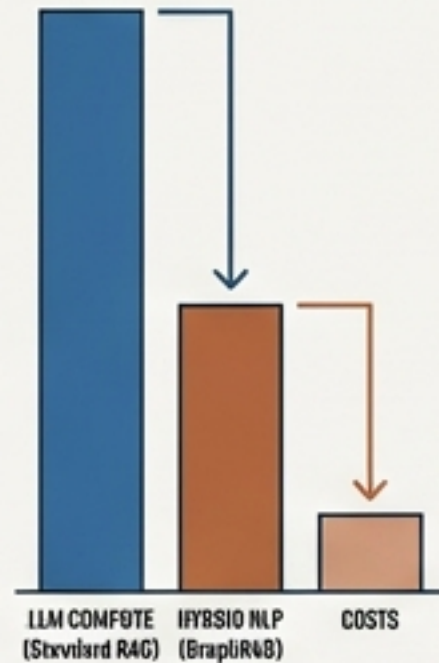
Zero-hallucination, transparent source tracing for strict regulatory compliance.



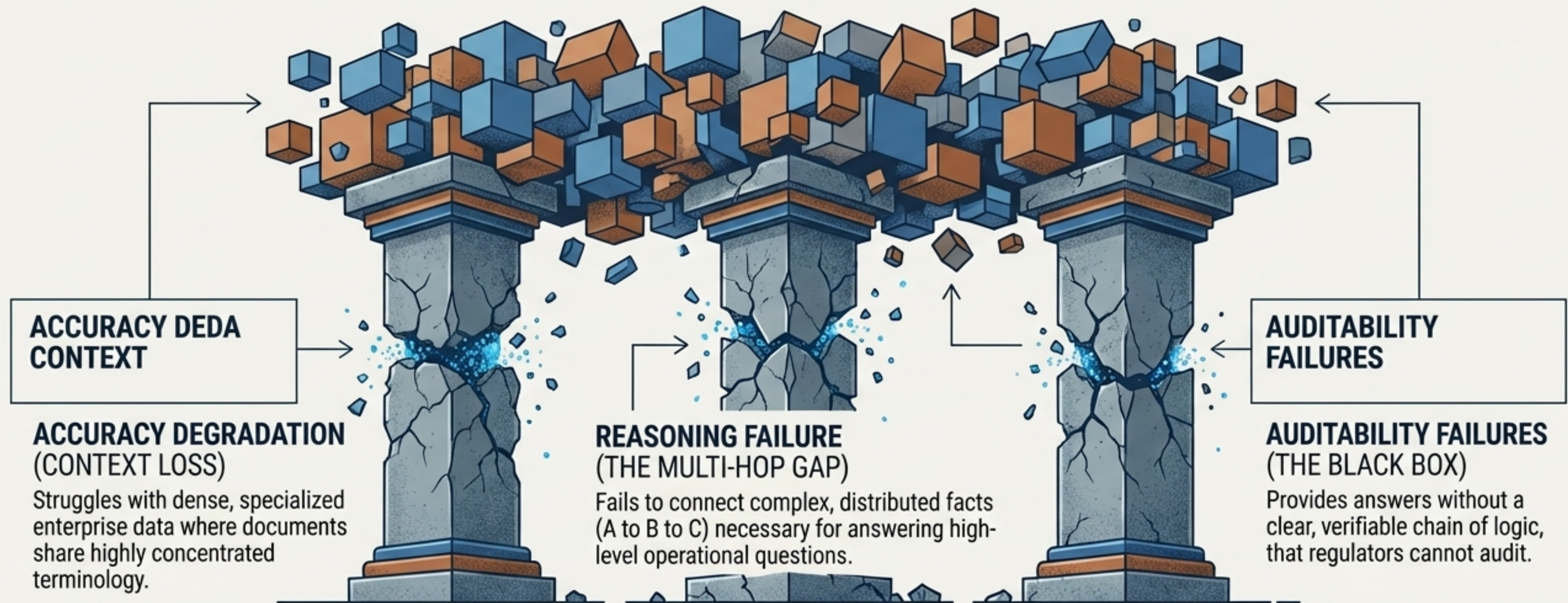
90%

COST REDUCTION

Using hybrid classical NLP extraction to bypass massive LLM compute costs at scale.



THE AI GROWTH CEILING: STANDARD RAG IS HITTING A WALL



THE ROOT CAUSE: THE CHUNKING PROBLEM

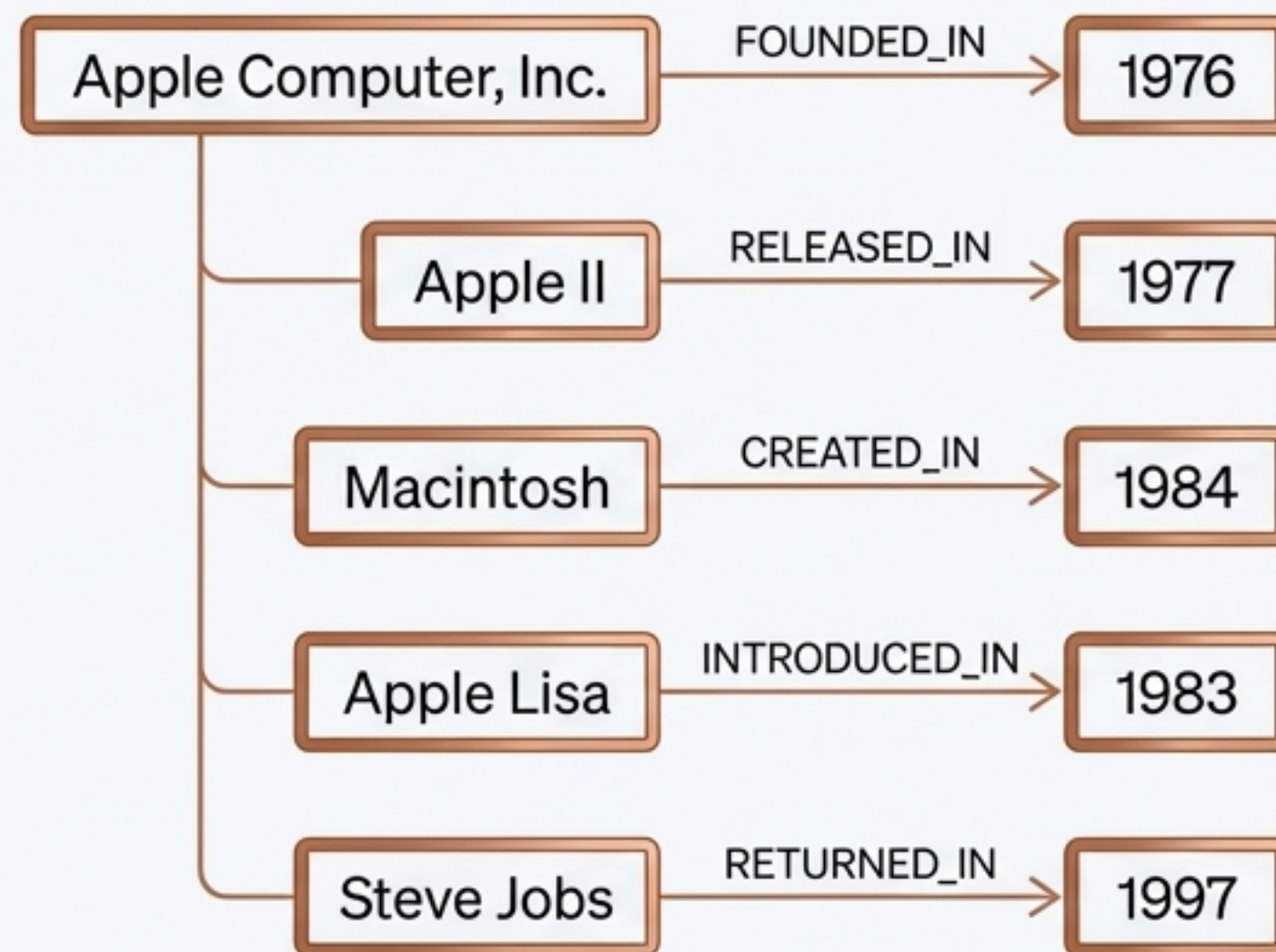
NAIVE CHUNKING

Apple Computer, Inc. was founded in 1976. The Apple II, released in 1977, was a major success. The Apple Lisa, introduced in 1983, was one of the first personal computers with a GUI. The Macintosh created in 1984, became iconic. 1984.

Steve Jobs returned in 1997, leading a period of innovation.

Slices enterprise knowledge into rigid blocks, creating proximity bias and destroying structural relationships (e.g., inaccurately attributing the Macintosh creation to 1983).

RELATIONAL PRESERVATION



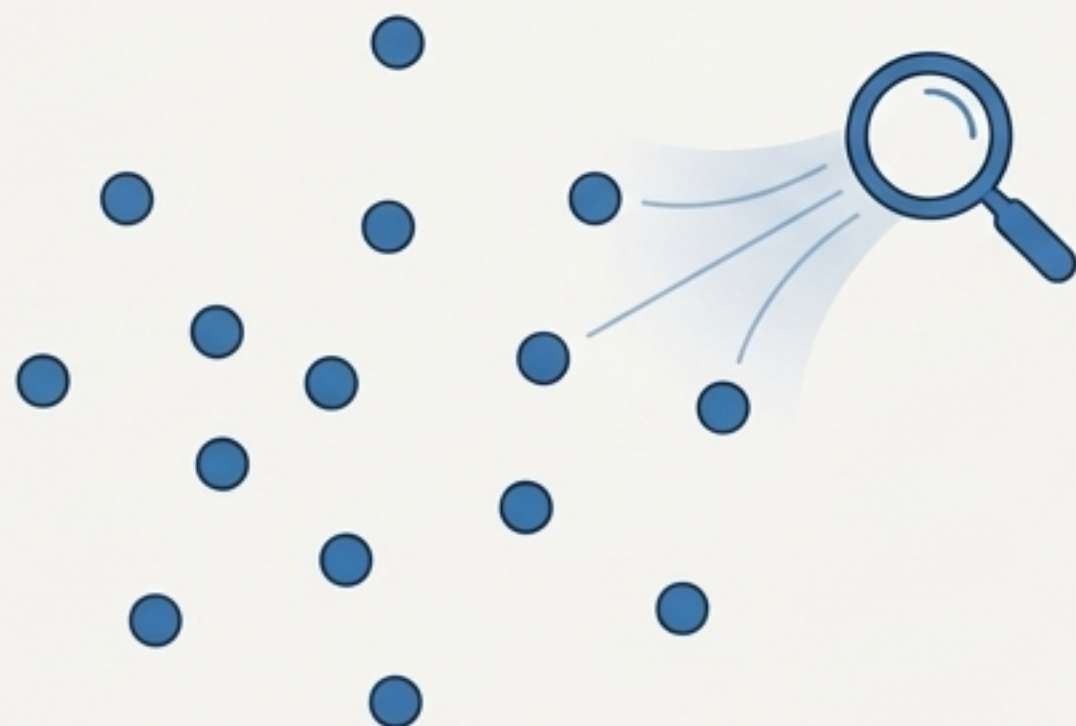
Graph structures capture the actual logic and conditions of the text, preserving the whole thought intact regardless of document length.

THE PARADIGM SHIFT: FROM STATIC VECTORS TO DYNAMIC GRAPHS

Key Takeaway: We are upgrading our AI from a simple matching engine into a reasoning engine.

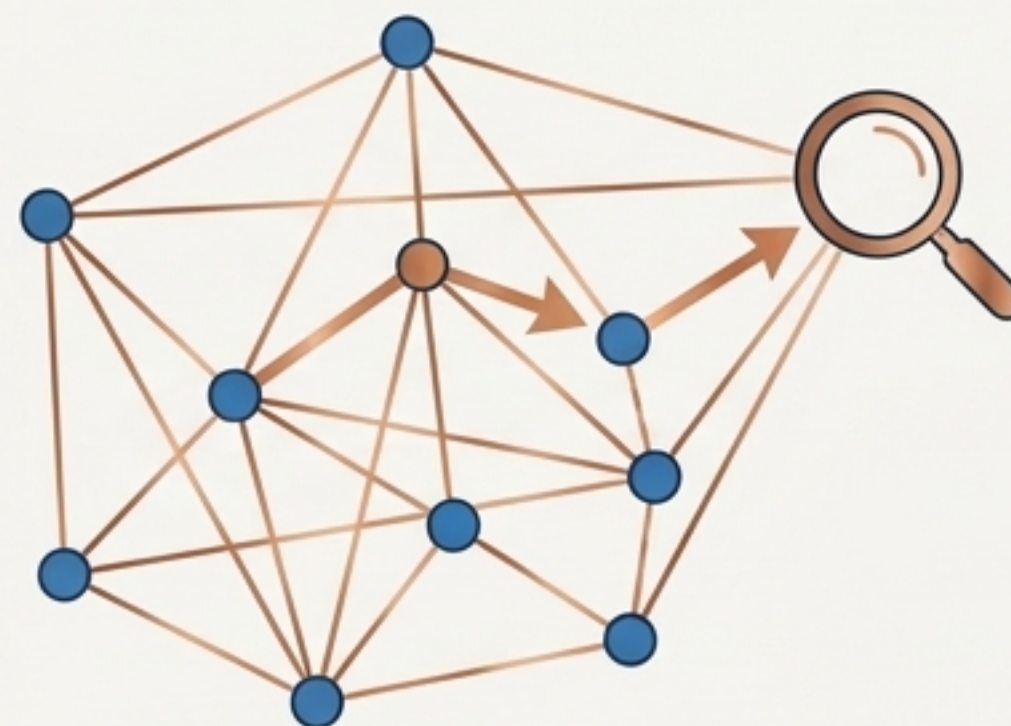
VECTOR SEARCH

Querying for isolated words that look statistically similar. (Blind proximity).

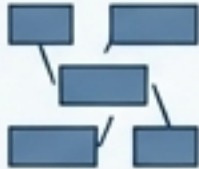
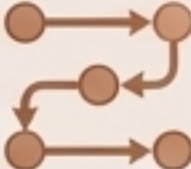
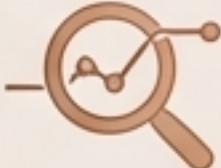


GRAPH RETRIEVAL

Querying an intelligent, interconnected map of relationships and keys. (Semantic logic).

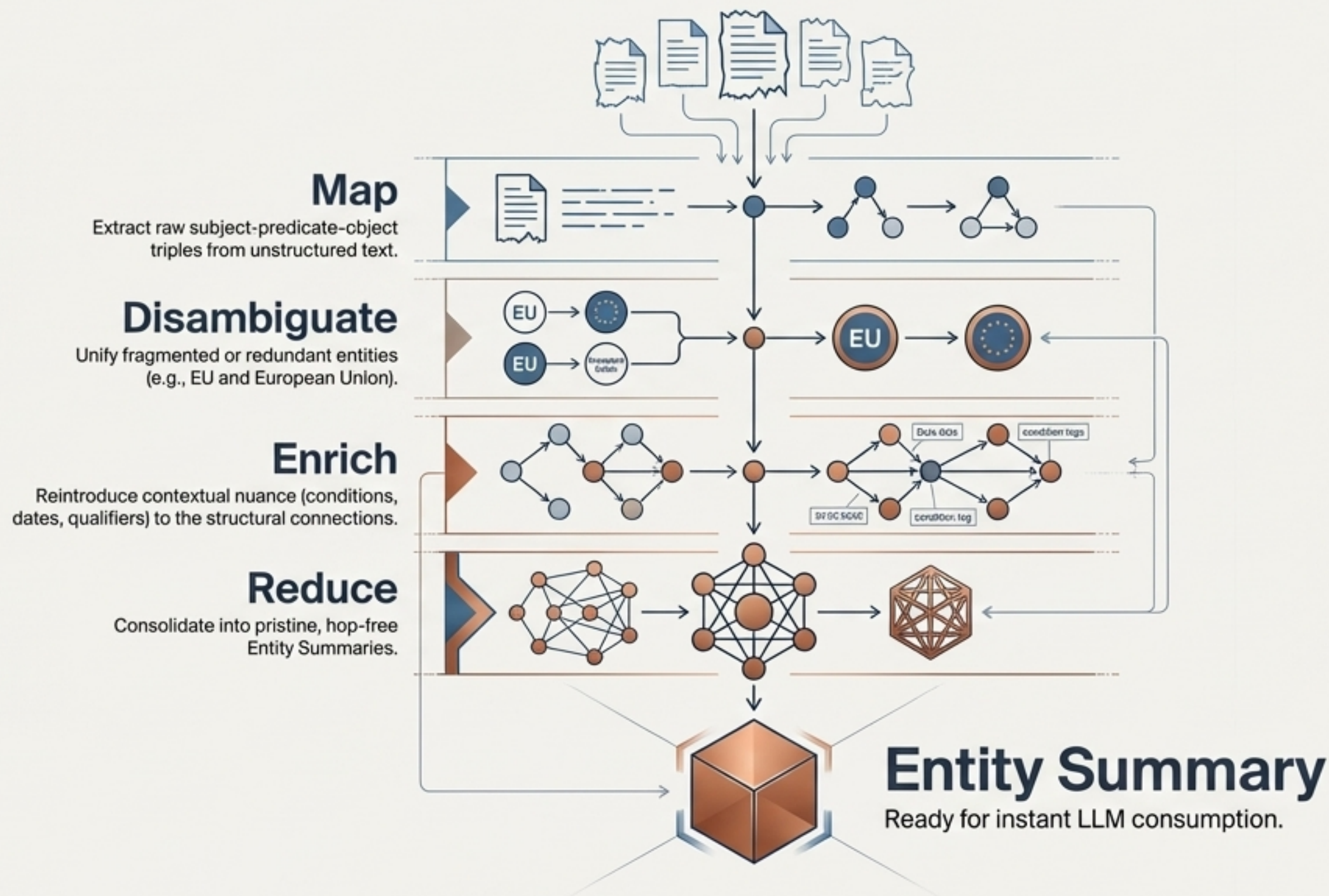


ENTERPRISE DIAGNOSTIC: VECTOR RAG VS. GRAPH RAG

EVALUATION DIMENSIONS	VECTOR RAG	GRAPH RAG
Data Handling	 Flat, isolated document chunks	 Interconnected entity networks
Complex Queries	 Fails at multi-hop synthesis	 Excels at multi-hop reasoning and composition
Auditability & Compliance	 Black Box (opaque origins)	 Fully transparent (100% source-traced relationships)
Enterprise Readiness	 High hallucination risk on dense data	 High fidelity, condition-aware precision

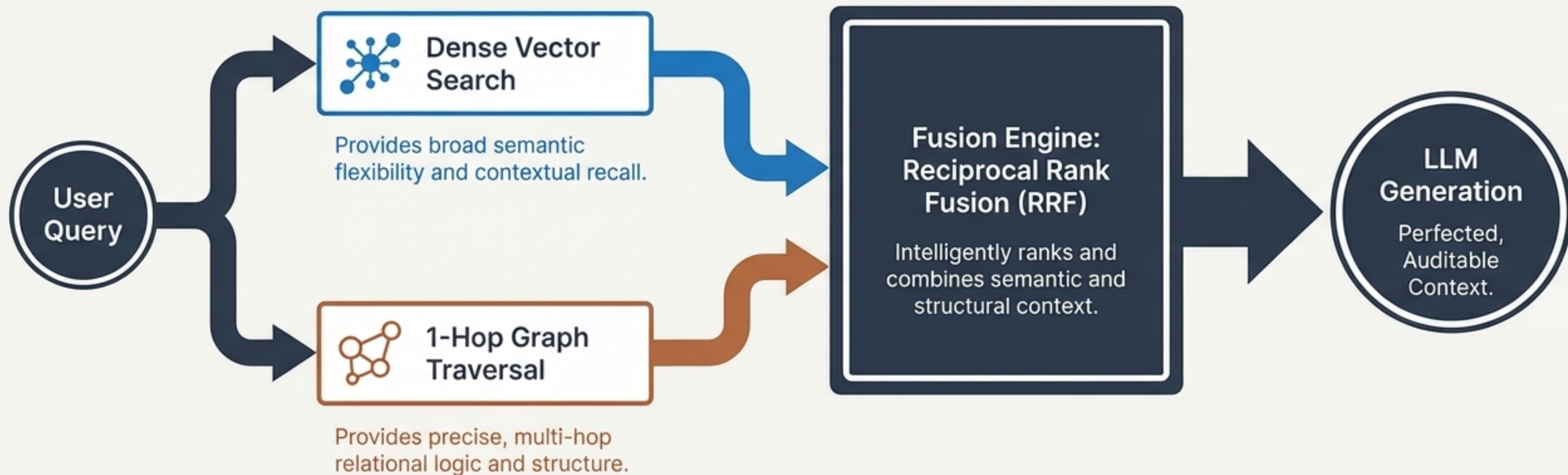
How It Works: The MDER Refinement Engine

Shift the Workload: By compressing multi-hop evidence into entity-centric summaries during indexing, we eliminate slow traversal during retrieval.



The Best of Both Worlds: Hybrid Retrieval Architecture

A cascaded retrieval strategy that guarantees high recall without sacrificing accuracy.

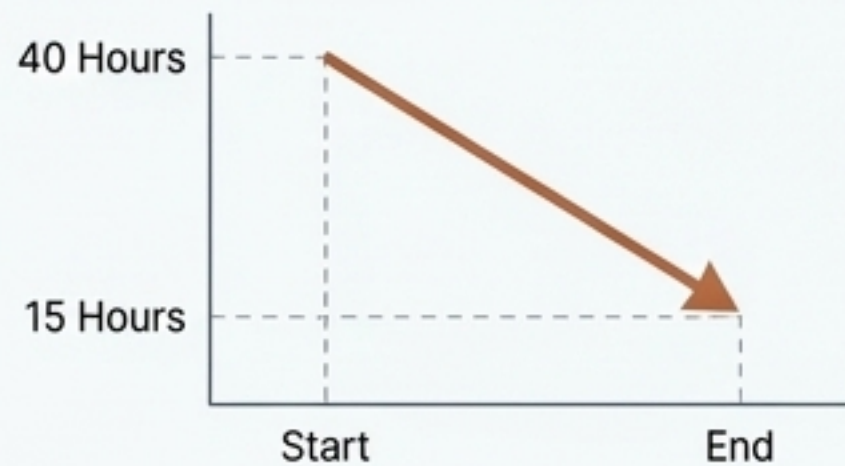


Industry Outcomes: The ROI of GraphRAG is Proven

28.6%

Drop in Resolution Time

LinkedIn: Reduced mean customer service issue resolution from 40 hours to 15 hours by parsing tickets into intra-issue trees and inter-issue graphs.



1st Place

Benchmark Accuracy

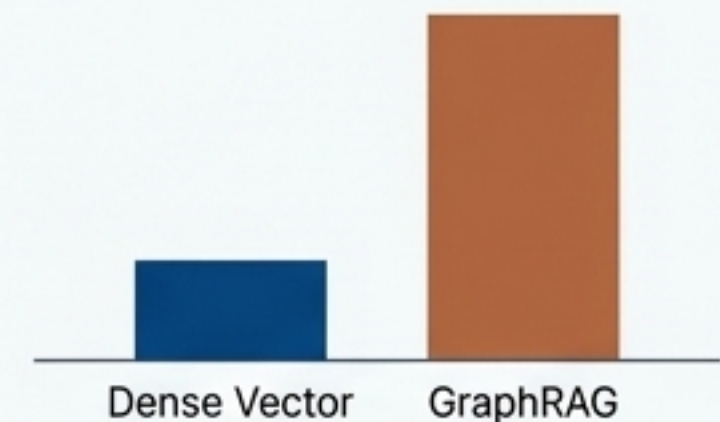
Writer: Defeated 7 enterprise vector systems. Enabled AI in regulated healthcare/finance by exposing exact reasoning snippets and ensuring zero-hallucination auditability.



78.5%

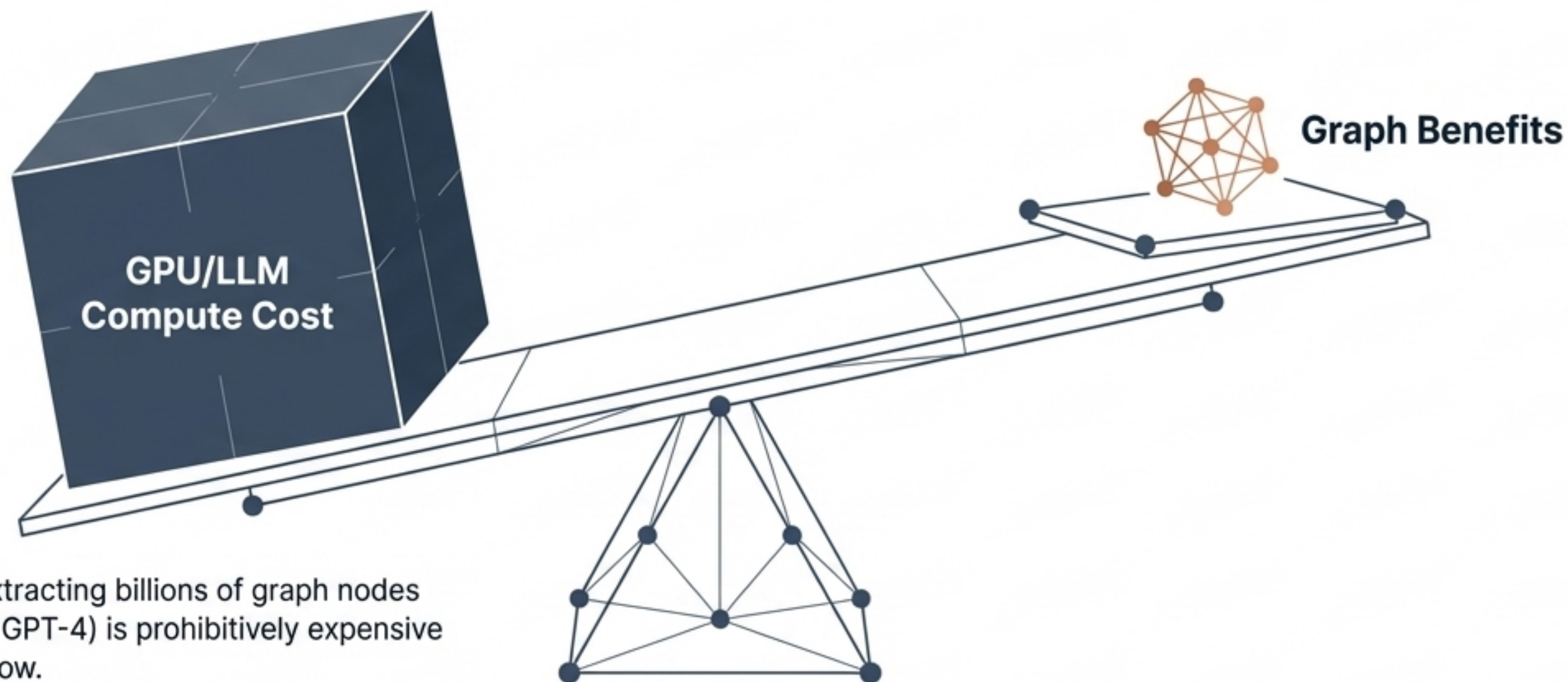
Win-Rate

SAP: Overpowered dense vector retrieval in complex legacy code migration (ABAP to S/4HANA) by understanding transaction-to-screen structural dependencies.



The Elephant in the Room: Cost & Scalability

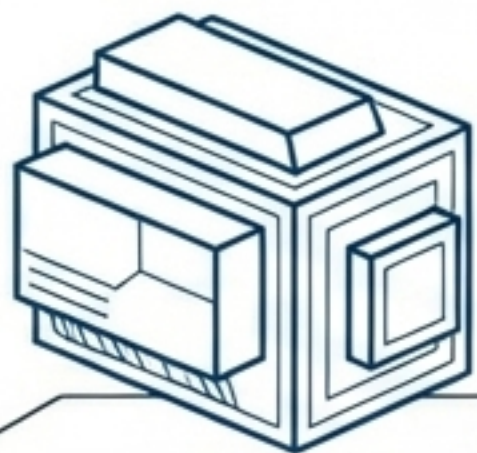
Standard GraphRAG architectures trade retrieval accuracy for unsustainable indexing costs.



- **The Problem:** Extracting billions of graph nodes using LLMs (like GPT-4) is prohibitively expensive and incredibly slow.
- **The Bottleneck:** Traditional graph databases choke on complex traversal operations at query time as enterprise data scales into terabytes.

Our Pragmatic, Scalable Intervention

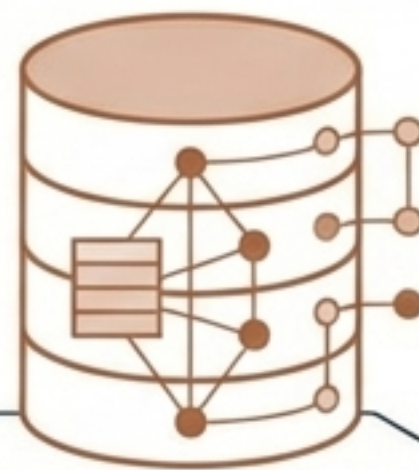
Classical NLP Engine



1. Cost-Efficient Extraction (The SAP Method):

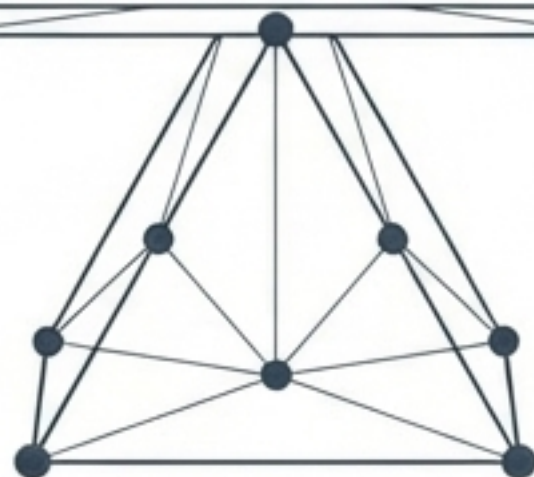
We deploy classical NLP (Dependency Parsing) for knowledge extraction instead of expensive LLMs.

Hybrid JSON Storage

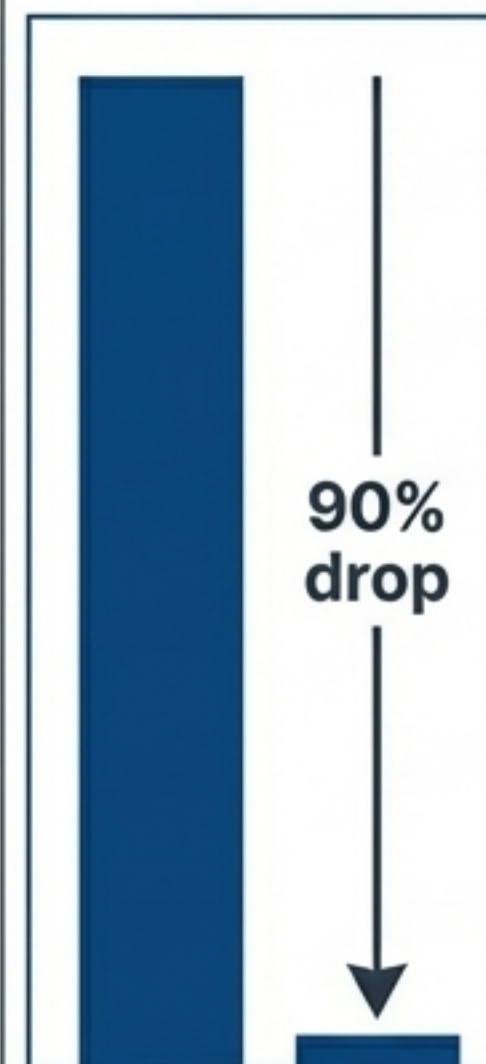


2. Hybrid Storage (The Writer Method):

We store graph data points as JSON in fast, existing Lucene-based search engines, preventing scaling bottlenecks.



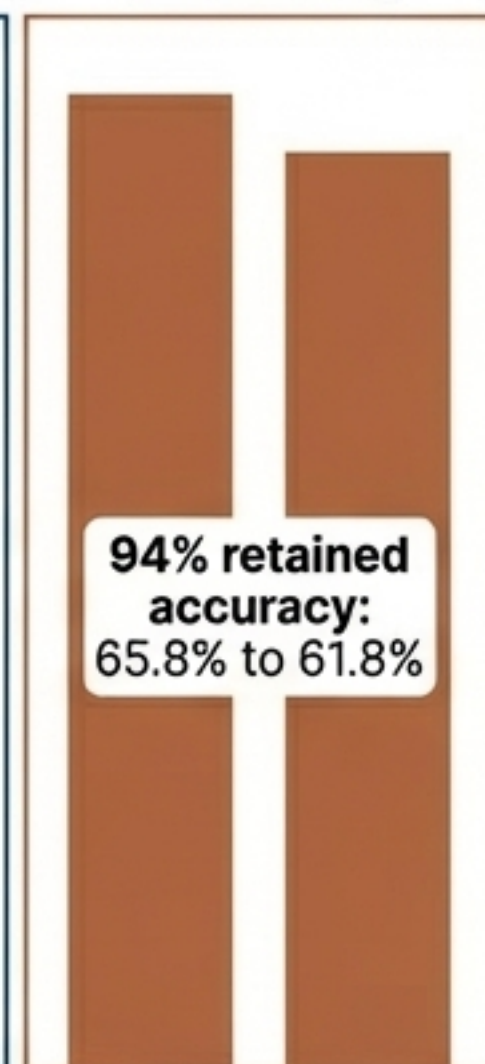
Cost



LLM Method

NLP Method

Accuracy



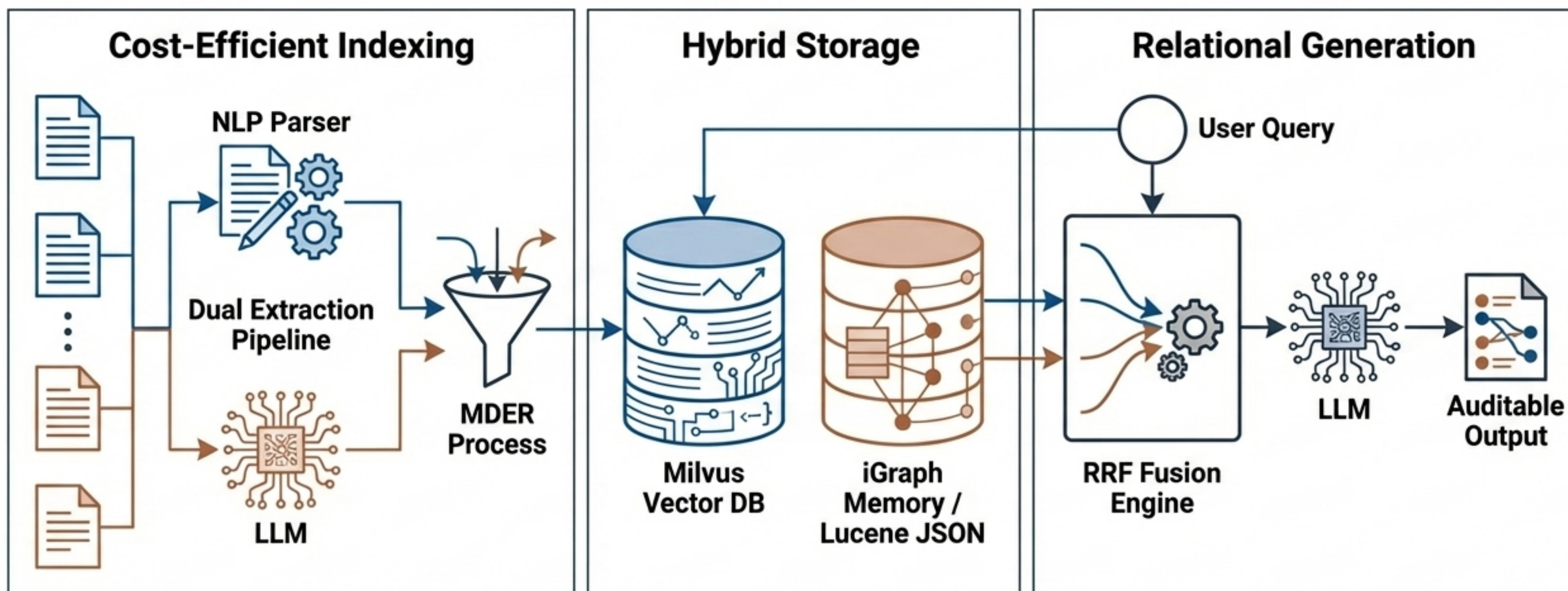
94% retained accuracy:
65.8% to 61.8%

LLM Method (GraphRAG)

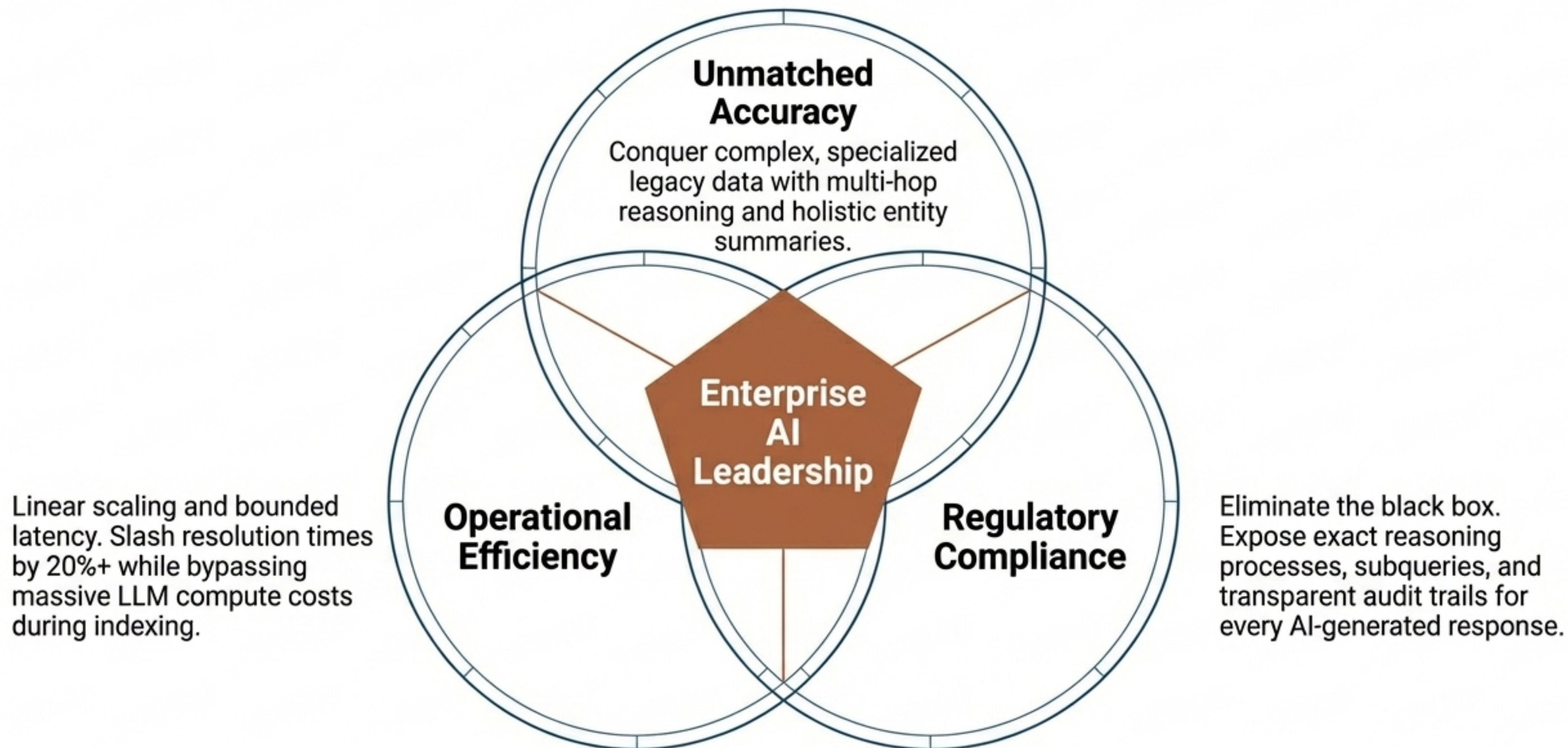
NLP Method (Classical NLP)

Blueprint: Our Target Enterprise Architecture

An end-to-end, production-ready pipeline that balances theoretical ideals with operational realities. Built for scale, engineered for efficiency.



Strategic Business Outcomes



The Ask: A 90-Day Sprint to ROI

The Request: Executive sponsorship and initial seed funding to build the MVP of our hybrid GraphRAG pipeline.



Phase 1: Foundation (Days 1-30)

Build the Dependency Parsing (NLP) extraction pipeline to ensure cost-efficiency from day one.

Phase 2: MVP Pipeline (Days 30-60)

Deploy the hybrid storage and Reciprocal Rank Fusion (RRF) retrieval engine.

Phase 3: Pilot Readout (Days 60-90)

Run a live pilot on our most complex legacy dataset (e.g., code migration or customer service tickets) and deliver a hard ROI readout.

The background of the slide features a complex network graph pattern. It consists of numerous small, light-brown circular nodes connected by thin, light-brown lines, creating a dense, interconnected web that fills the entire background.

The Future of Enterprise AI is Connected

Data without relationships is just noise. By shifting from static vectors to dynamic graphs, we are not just upgrading an algorithm—we are building an auditable, reasoning brain for the entire enterprise.

Ready for Strategic Deployment.